

Figure 1.1 – The components of the Databricks Data Intelligence Platform

Let's check out the following list with the descriptions of the items in the figure:

- **Delta Lake**: The data layout within the Data Intelligence Platform is automatically optimized based on common data usage patterns
- **Unity Catalog**: A unified governance model to secure, manage, and share your data assets
- **Data Intelligence Engine**: This uses AI to enhance the platform's capabilities
- **Databricks AI**: ML tools to support end-to-end ML solutions and **generative AI** capabilities, including creating, tuning, and serving LLMs
- **Delta live tables**: Enables automated data ingestion and **data quality**
- **Workflows**: A fully integrated orchestration service to automate, manage, and monitor multi-task workloads, queries, and pipelines
- **Databricks SQL (DBSQL)**: An SQL-first interface, similar to how you would interact with a data warehouse, and with functionality such as text-to-SQL, which lets you use natural language to generate queries

Now that we have our elements defined, let's discuss how they help us achieve our ML goals.

Advantages of the Databricks Platform

Databricks' implementation of a lakehouse architecture is unique. Databricks' foundation is built on a Delta-formatted data lake that Unity Catalog governs. Therefore, it combines a data lake's scalability and cost-effectiveness with a data warehouse's governance. This means not only are table-level permissions managed through **access control lists** (**ACLs**) but file and object-level access are also regulated. This change in architecture from a data lake and/or a data warehouse to a unified platform is ideal – a lakehouse facilitates a wide range of new use cases for analytics, business intelligence, and data science projects across an organization. See the *Introduction to Data Lakes* blog post in the *Further reading* section for more information on lakehouse benefits.